



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

Program/s: B. Tech/MBA Tech

Year: II Semester: IV

Stream/s B.Tech AI & DS/ B.Tech AI & ML/MBA Tech AI

Subject: Machine Learning

Time: 3 hrs (10 AM to 1 PM)

Date: 27/06/2023

No. of Pages:4

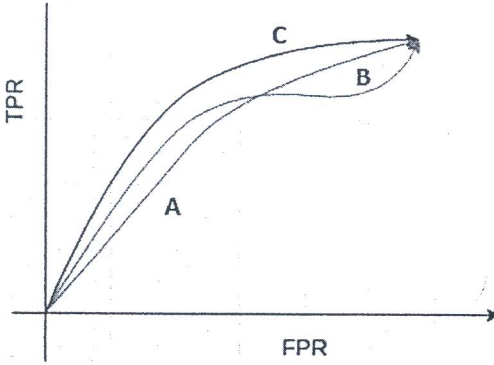
Marks: 100

Re- Examination

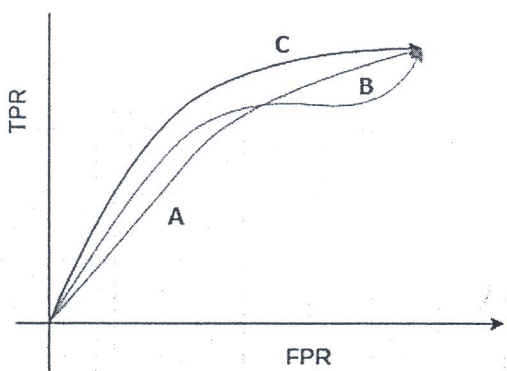
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]												
CO-1; SO- 1; BL- 3	a.	Assume you have a data set that summarizes a marketing campaign with information related to prospective customers. The data set contains 100 observations with several columns that summarize information about the prospective customer. It also has a column that flags whether the prospect responded or not. -Identify the appropriate category of machine learning algorithm can be applied to above task? and specify reason why you select that one. -Write any one algorithm name that can be applied to above task -What kinds of output labels can we make based on the above task? -What are the different performance measures can be used for evaluation of above task?													
CO- 1;SO- 2; BL-3	b.	Apply the IQR method on following data and identify the outliers. 0, 0, 2, 5, 8, 8, 8, 9, 9, 10, 10, 10, 11, 12, 12, 12, 14, 15, 20, 25.													
CO-2 ;SO-2 ; BL-2	c.	For below data calculate Coefficient of Determination for simple linear regression and state your inference. The regression line for the same is $1.60x+3.40$. <table border="1"><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Y</td><td>5</td><td>7</td><td>8</td><td>9</td><td>12</td></tr></table>	X	1	2	3	4	5	Y	5	7	8	9	12	
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CO-3; SO- 1; BL-2	d.	Q1. The ROC plots of three classifiers is shown in following figure  What is ROC curve? Which is the best classifier? Justify your answer.																			
Q2 CO-1; SO-1; BL-1	a.	Why do we need dimensionality reduction? What are its drawbacks?	[10]																		
Q2 CO-1; SO-1; BL-2	b.	What are the various categorical feature types and describe the various encoding methods for categorical data?	[10]																		
Q3 CO-2; SO-1; BL-3	a.	Justify the statement “Underfitted models have high bias and Overfitted models have high variance”. Illustrate and explain the working of 3-fold cross validation method.	[10]																		
Q3 CO-2; SO-2; BL-4	b.	A study was conducted to understand the effect of number of hours the students spent studying to their performance in the final exams. You are given the following 8 samples from the study. What is the best linear fit on this dataset? <table><thead><tr><th>No of hours spent studying (x)</th><th>Score in the final exam (0-100)(y)</th></tr></thead><tbody><tr><td>10</td><td>95</td></tr><tr><td>9</td><td>80</td></tr><tr><td>2</td><td>10</td></tr><tr><td>15</td><td>50</td></tr><tr><td>10</td><td>45</td></tr><tr><td>16</td><td>98</td></tr><tr><td>11</td><td>38</td></tr><tr><td>16</td><td>93</td></tr></tbody></table>	No of hours spent studying (x)	Score in the final exam (0-100)(y)	10	95	9	80	2	10	15	50	10	45	16	98	11	38	16	93	[10]
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Q4 CO-3; SO-2; BL-4	a.	What is the important criteria to apply Naïve Bayes’ algorithm. Consider the cars dataset having attributes color, type, origin, and the target as Stolen (yes or no). Predict the possibility of a Red Domestic SUV to be stolen using Naïve Bayes classifier.	[10]																		

			Color	Type	Origin	Stolen?	
		1	Red	Sports	Domestic	Yes	
		2	Red	Sports	Domestic	No	
		3	Red	Sports	Domestic	Yes	
		4	Yellow	Sports	Domestic	No	
		5	Yellow	Sports	Imported	Yes	
		6	Yellow	SUV	Imported	No	
		7	Yellow	SUV	Imported	Yes	
		8	Yellow	SUV	Domestic	No	
		9	Red	SUV	Imported	No	
		10	Red	Sports	Imported	Yes	
Q4 CO-3; SO-2; BL-3	b.	The classifier's performance needs to be computed. The ground truth and actual label are given below y_true = ['P', 'N', 'N', 'N', 'P', 'P', 'N'] y_pred= ['P', 'N', 'N', 'P', 'P', 'N', 'N'] i. Compute the confusion Matrix and find the accuracy of the classifier. ii. Complete the classification report given below					[10]
		Class	Precision	Recall	F1 score		
		N					
		P					
Q5 CO-3; SO-1; BL-2	a.	What do you understand by random forest algorithm? What are its benefits? What are the steps involved in random forest algorithm? Explain with the help of an example					[4] [6]
CO-3; SO-1; BL-2	b.	What are the similarities and dissimilarities in bagging and boosting techniques? What are disadvantages of using ensemble learning?					[10]
Q6 CO-3; SO-1; BL-2	a.	What do you understand by K-Means clustering? What are its advantages and disadvantages? Explain the steps involved in finding clusters using K-Means algorithm.					[10]
CO-3; SO-1; BL-2	b.	What do you understand by DBSCAN? Explain utility of DBSCAN and types of data points used. Write down the steps used in DBSCAN algorithm.					[10]
Q7 CO-1; SO-1; BL-2	a.	Explain different techniques used for feature scaling					[5]
Q7 CO-3; SO-1; BL-3	b.	Illustrate the working of logistic Regression					[5]
Q7	c.	Explain XGBoost. What are the benefits and attributes of XGBoost?					[5]

CO-3; SO- 1; BL-2	d.	Q1. The ROC plots of three classifiers is shown in following figure  What is ROC curve? Which is the best classifier? Justify your answer.																			
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Q7 CO-3; SO-1; BL-2	d.	What do you understand by divisive/top-down clustering? Write down the steps involved in divisive clustering algorithm.	[5]